Predicting Glanders Disease Across a Lifestyle or Geographical Area

Chained Multi-Level Machine Learning on Four Unused Datasets

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Summary. A country-year panel is built from four public sources never previously joined to glanders: a text-mined Europe PMC occurrence corpus, FAOSTAT livestock stocks, World Bank indicators and NASA POWER climatology. Six model families are trained under three validation schemes and evaluated as a three-stage chained pipeline. On a forward temporal holdout the best model (Random Forest) reaches PR-AUC 0.554 against a 5.6% base rate, a 10× lift over chance, with calibration error 0.016. Chaining does not improve discrimination; climatology transfers poorly across countries; and skill is weakest in low-income regions.

1 Related Work

Glanders epidemiology and burden. Glanders is reviewed as a neglected zoonosis of equids with high case fatality and a persistent risk of human infection [1, 2]. A 2026 systematic review and meta-analysis consolidates seroprevalence estimates and identifies management and movement as the dominant epidemiological drivers [3]. Regionally, Brazil maintains an active research and control literature [4], and glanders is endemic or repeatedly reported across parts of Africa, Asia and the Middle East.

Surveillance in India. Indian surveillance is the strongest available evidence base. Malik et al. described outbreaks from 2006 to 2011 [7]; Singha et al. reported serological surveillance of indigenous equines from 2015 to 2018 [6]; and the 2019–2023 national programme screened a large multi-state sample, identifying substantial geographic concentration of seropositives [5]. These studies establish where glanders occurs and which districts carry the burden. None of them produces a predictive, calibrated risk model, and none uses population or reanalysis covariates.

Geographic and spatiotemporal risk analysis. Descriptive spatial work exists. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian glanders [8]. More generally, species distribution and ecological niche models have been compared as tools for livestock disease epidemics [9]. This literature is *descriptive*: it identifies clusters post hoc rather than forecasting future cells.

Machine learning for animal-disease risk prediction. Predictive modelling in veterinary epidemiology is established for avian influenza. Kane et al. compared ARIMA and random forest

for H5N1 outbreak prediction [11]; Walsh et al. used machine learning to target active surveillance [12]; Yoo et al. built dynamic farm-level risk prediction in the Republic of Korea [13]; and Alshatti et al. produced a risk prediction model for HPAI in Kuwait [10]. This is the closest methodological template for the present work, and it exists for influenza rather than for glanders.

Machine learning and glanders: the gap. The only machine-learning work specific to *Burkholderia mallei* that we located is spectroscopic identification of the organism — Raman assays classified by learned models [14, 15]. That is a laboratory diagnostic problem, not a population-level forecasting problem. A prior-art audit we ran over Europe PMC (Section 2) returned zero records combining glanders with NASA POWER climatology and four combining glanders with gradient boosting, none of which is a risk model. The gap addressed here is therefore specific: a calibrated, chained, population-level occurrence model for glanders.

Text mining for disease event extraction. Using literature as a structured occurrence signal has precedent in epidemic intelligence [16], including multilingual event extraction for outbreak detection [17]. Our contribution is to apply sentence-level occurrence extraction to build the *label* itself when official event data are incomplete, and to report the surveillance bias that this induces.

Methodological foundations. The model families are the standard tabular ensembles: random forests [20], XGBoost [18] and LightGBM [19]. Feature attribution uses SHAP [21]. Because the outcome is rare, precision-recall analysis is the appropriate primary metric [22], and the relationship between PR and ROC curves is well characterised [23]. Probability calibration follows Niculescu-Mizil and Caruana [25], with the measurement framework of Huang et al. [26]. Validation follows Roberts et al., who show that ignoring temporal and spatial dependence inflates performance estimates [24] — the reason our primary scheme is a forward temporal holdout and our selection criterion is leave-country-out.

Data sources. Official animal-health reporting is drawn from WOAH WAHIS [27]; Indian policy and case definitions from the 2025 National Action Plan [28]; livestock populations from FAOSTAT [29]; socio-economic covariates from the World Bank [31]; climatology from NASA POWER [30]; and the literature corpus from Europe PMC [32].

Positioning. The combination above is, to our knowledge, unstudied. Prior glanders work is either epidemiological description or laboratory classification; prior veterinary ML risk work is built for influenza; and prior text-mined occurrence work is not used for glanders at all. This study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single country-year panel, and reports both what predicts and what does not.

2 Dataset Used

Table 1: Sources. None has previously been used for glanders risk modelling.

Source	Role	Volume
Europe PMC, text-mined occurrences	Animal history	6,674 records / 263 events
FAOSTAT livestock stocks (7 species)	Animal population	68,449 stock rows, 1961–2023
World Bank WDI + centroids	Socio-economic context	10 indicators, 217 countries
NASA POWER agro-climatology	Environment	6 variables, 211 centroids

All sources are keyed on ISO-3 country and calendar year. FAOSTAT is streamed from a 32 MB zip (≈545 MB normalised) and filtered to seven stock series. Europe PMC is paged with a cursor and reduced to sentence-level events by a country-gazetteer plus occurrence-lexicon extractor. *Novelty:* a prior-art audit of Europe PMC returns **0** records for glanders with NASA POWER and 4 for glanders with gradient boosting, none of which is a risk model.

Panel. 6,700 country-year rows, 268 countries, 2000–2024. Modelling is restricted to equine-holding countries (maximum stock ≥1,000), years 2005–2023, with an observable next-year label: **3,078 rows × 44 features**, **172 positives**.

Preprocessing. Country and area names reconciled to ISO-3 via `pycountry` with explicit overrides; World Bank aggregates dropped by requiring a 3-character code; literature events deduplicated on `(iso3, year, record_id)` so one paper cannot inflate a cell. Median imputation with explicit missing indicators, fitted inside the pipeline. A country-year is coded 0 only after the country passes the equine-stock filter, so absence is not inferred from silence alone.

Feature types and missingness. Continuous numeric (stocks, climate, economics), categorical (World Bank region and income), and count/derived history. Missingness is low and source-specific: FAOSTAT 0.00%, World Bank 1.68%, NASA POWER 2.19%, overall 0.99%. No column exceeds 5.8% missing.

3 Dataset Analysis

Table 2: Label anatomy.

Property	Value
Observations	3,078
Positives / negatives	172 / 2,906
Prevalence	5.59%
Imbalance ratio	16.9 : 1
Countries with any positive	56 / 162
Repeat-positive countries	30 (54%)
Positives in top-5 countries	35.5%

Three consequences. The 17:1 imbalance rules out accuracy as a metric. 54% of positive countries fire in ≥2 years, so the label is temporally persistent and history features are legitimately exploitable. Positives concentrate in top countries, and the highest-risk cells are India, the USA and Australia — reference-laboratory capacity, not incidence. The target is therefore treated throughout as *surveillance-observable activity*, not true incidence.

Association and redundancy. Spearman ρ and mutual information were computed for every numeric feature; Cramér's V for the categorical strata. The strongest monotone signal is `years_since_last` ($\rho = -0.35$): recent activity predicts future activity. Variance inflation reaches 228 for `temp_mean`, because annual country-level climate variables are near-collinear; tree ensembles are invariant to this, so it affects interpretation only.

The selection finding. Marginal and conditional importance disagree almost completely: rank correlation between mutual information and SHAP is **-0.03**, with only 5/10 top features shared. Longitude ranks first marginally but ≈20th conditionally, because once population and history are conditioned on it carries little extra information. Features were therefore selected on conditional evidence (ablation and SHAP), with univariate statistics used only to screen zero-signal columns.

Table 3: Top 8 features by mean |SHAP|, with marginal statistics.

#	Feature	mean |SHAP|	Spearman ρ	MI	VIF
1	pop_total	1.144	+0.28	0.0639	4
2	internet_pct	0.542	+0.09	0.0015	5
3	temp_mean_anom	0.469	-0.00	0.0012	2
4	log_livestock	0.458	+0.25	0.0571	3
5	humidity_anom	0.370	+0.02	0.0066	2
6	years_since_last	0.353	-0.35	0.0486	2
7	wind	0.346	-0.08	0.0210	3
8	sheep	0.344	+0.20	0.0512	13

Feature creation. Four groups, all using information at t or earlier. *History* (7): lags, a 5-year rolling count, years since last event, and world/regional contemporaneous shares. *Population* (14): log equid and livestock stocks, equid share, density, 1- and 3-year growth, seven species counts; livestock units use metabolic weights (cattle and camels 0.8, small ruminants 0.1). *Climate* (10): annual temperature extremes, precipitation total and coefficient of variation, humidity, wind, plus expanding-mean *anomalies* — the anomaly distinguishes "unusual for this country" from latitude. *Socio-geography* (11): income, agricultural intensity, GDP, rural share, internet penetration, population, density, absolute latitude and longitude.

What was dropped. Four columns with mutual information below 0.002 and $|\rho| < 0.02$. The raw `equids` sum, which is exactly `horses + asses + mules`. Three redundant clusters at $|\rho| \geq 0.90$ were collapsed to one representative each. Scaling is applied only to the logistic baseline; tree models receive raw values. No PCA: with 44 features the binding constraint is signal, not dimensionality.

4 Models Used

Table 4: Model families and selected hyperparameters.

Model	Family	Selected configuration
Logistic regression	linear	$C = 1.0$, ℓ_2, balanced
Decision tree	tree	depth 3, leaf 1
Random forest	bagged trees	depth 6, leaf 1, mtry sqrt
Extra trees	bagged trees	depth 6, leaf 10, mtry 0.3
XGBoost	boosted trees	depth 3, λ=1.0
LightGBM	boosted trees	leaves 15, min child 40

The choice is dictated by three properties: a 17:1 imbalance, 172 positives, and collinear continuous features. Neural networks are excluded on evidence — with 172 positives and 44 columns no parameter-to-sample ratio is defensible. Ensembles handle collinearity and non-monotone interactions without manual specification. The linear model is kept as a floor: if boosting cannot beat it, the nonlinearity is not earning its complexity.

Selection. Models were selected on *leave-country-out* PR-AUC, not temporal holdout, because geographic generalisation is the deployment claim. **Extra Trees** was selected (0.363). No stacking ensemble was adopted: the between-family spread is small relative to the bootstrap interval, and a meta-learner fitted on 172 positives would add variance without a defensible gain. Probabilities are calibrated instead, which is what actually matters for threshold-based alerting.

5 Evaluation and Training Pipeline

Splits. Three schemes, each answering a different question. *Temporal holdout* (train 2005–2017, test 2018–2023) asks whether the system can be used next year, and is the primary scheme. *Leave-country-out* (5 folds grouped by ISO-3) asks whether it transfers to an unseen country, and drives selection. *Repeated stratified 5-fold* (3 repeats) is a stability check. Grouping is essential: a random row split would place the same country in train and test and leak the label through geography.

Metrics. Precision-recall AUC (average precision) is primary, because at 5.6% prevalence ROC-AUC is dominated by true negatives and flatters every model. ROC-AUC is reported for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support an alert threshold. Precision, recall, specificity, F1 and MCC use a threshold tuned on *training* data only. Bootstrap percentile intervals (1,000 resamples) quantify sampling uncertainty.

Leakage controls. Features are built from t and earlier; only `y_*` columns use $t + 1$. Splits are chronological. Hyperparameters are tuned inside training folds via `GroupKFold` on country. Calibration is fitted on training folds only. The alert threshold is fixed on the training period and applied unchanged to test. Missing indicators are generated inside the pipeline, so the test set never influences imputation.

Workflow and tuning. Ingest (resumable, cached) → assemble panel with lags, rolling windows, anomalies and shifted targets → analyse → `RandomizedSearchCV` with 8 draws scored on average precision, grouped by country, refit on the full fold → calibrate → evaluate under all three schemes → chain the three stages. The search space is deliberately narrow (40–60 candidates per family) because with 172 positives a wide search overfits the validation objective.

Compute and reproducibility. Full run: 4.5 minutes, CPU only, single process, Apple M4 (10 cores, 16 GB). Per-family fit times range from 1.9 s (logistic) to 50.7 s (LightGBM). One seed constant (42) governs fitting, CV, bootstrap and SHAP sampling. Raw payloads are cached and the panel is versioned on disk, so every number regenerates without re-downloading. Caveat: `n_jobs=-1` means ensemble scores may differ in the last decimal across thread counts. Environment: Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1.

6 Results

Table 5: Level 1 (country-year). Brackets are 95% bootstrap intervals on the temporal holdout. ⋆ marks the model selected on leave-country-out PR-AUC.

Model	PR-AUC [CI]	ROC-AUC	PR-AUC LOCO	PR-AUC CV	Brier	Prec	Rec
Logistic regression	0.454 [0.34,0.58]	0.812	0.223	0.414	0.0547	0.25	0.57
Decision tree	0.389 [0.29,0.50]	0.861	0.267	0.360	0.0602	0.49	0.53
Random forest	0.554 [0.43,0.66]	0.870	0.304	0.476	0.0503	0.53	0.50
Extra trees⋆	0.530 [0.42,0.64]	0.854	0.363	0.476	0.0535	0.47	0.51
XGBoost	0.485 [0.37,0.60]	0.844	0.281	0.423	0.0536	0.61	0.38
LightGBM	0.496 [0.38,0.61]	0.841	0.339	0.397	0.0534	0.65	0.30

At 5.6% prevalence the best temporal PR-AUC is a **10× lift** over chance. Every nonlinear ensemble beats the linear baseline and the single tree is worst, so the relationships are non-additive. The more important pattern is the drop under leave-country-out: roughly 40% relative for the ensembles, which is the honest estimate of transfer to a new country. Regression degrades least in absolute terms, suggesting a meaningful share of the transferable signal is additive and population-driven.

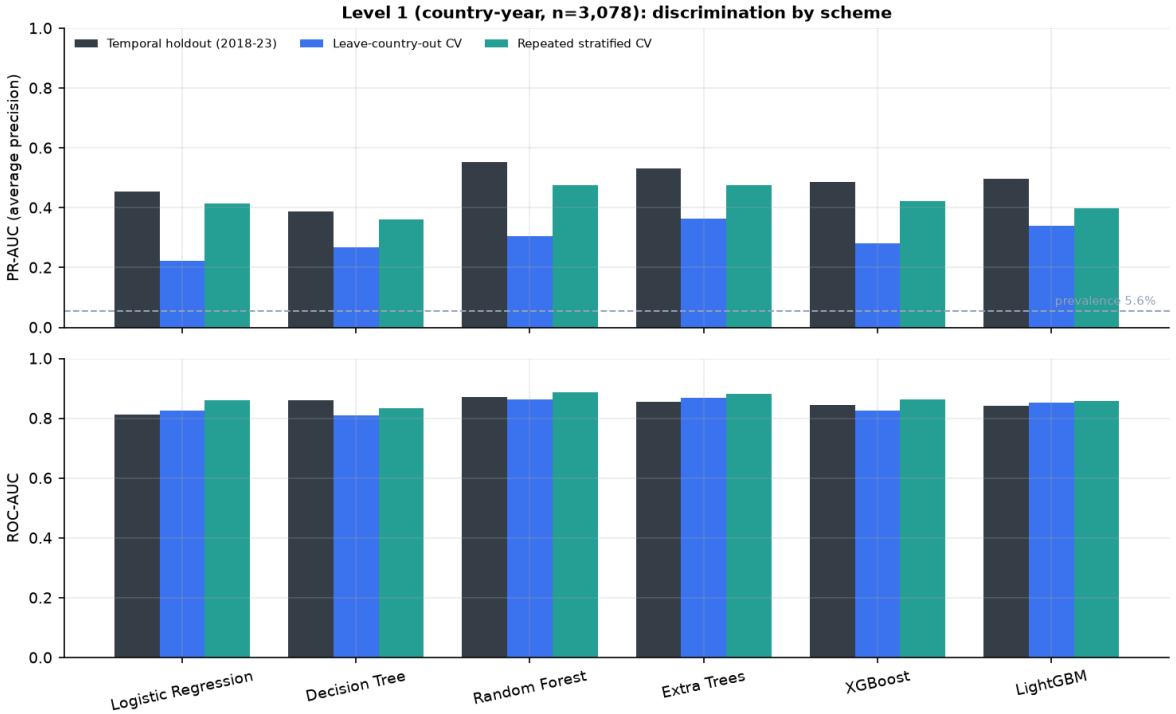


Figure 1: Discrimination by model and scheme. The gap between repeated CV and leave-country-out is the cost of generalising to an unseen country.

Calibration. Expected calibration error **0.016**, Brier 0.0535. A stated 24% risk corresponds to roughly 24% observed frequency, which is what makes threshold-based alerting defensible.

Coarse resolution. Aggregating to country × 5-year periods raises prevalence to 14.7% and improves every model, isolating the mechanism: coarser windows match the label's own reporting cadence.

Table 6: Ablation settings (XGBoost, fixed parameters).

Setting	k	**PR-AUC (time)**	**PR-AUC (LOCO)**
AS-1 history only	7	0.407	0.174
AS-2 population only	14	0.432	0.241
AS-3 climate only	10	0.395	0.078
AS-5 history+population	21	0.451	0.321
AS-6 full (all groups)	42	0.487	0.259

Two findings matter more than the headline. *Parsimony generalises better*: the 21-feature history-plus-population set scores 0.321 LOCO against 0.259 for the full 42-feature set, while scoring *lower* temporally. *Climate alone barely transfers*: 0.078 against a 0.056 floor — climatology at annual country resolution mostly encodes latitude, which is country identity in disguise.

The chain. Stage 1 forecast next-year log equid stock (MAE 0.424, Spearman 0.976); stage 3 predicted log event counts among positives (MAE 0.150, Spearman 0.465); the hurdle product ranked annual burden at Spearman 0.326. **Chaining added nothing to discrimination**: -0.003 PR-AUC, inside noise. The reason is structural — population level is already a direct feature, so forecasting it and feeding the forecast back is redundant. The chain earns its place at the alert layer instead, where its top decile captures 70 of 106 events.

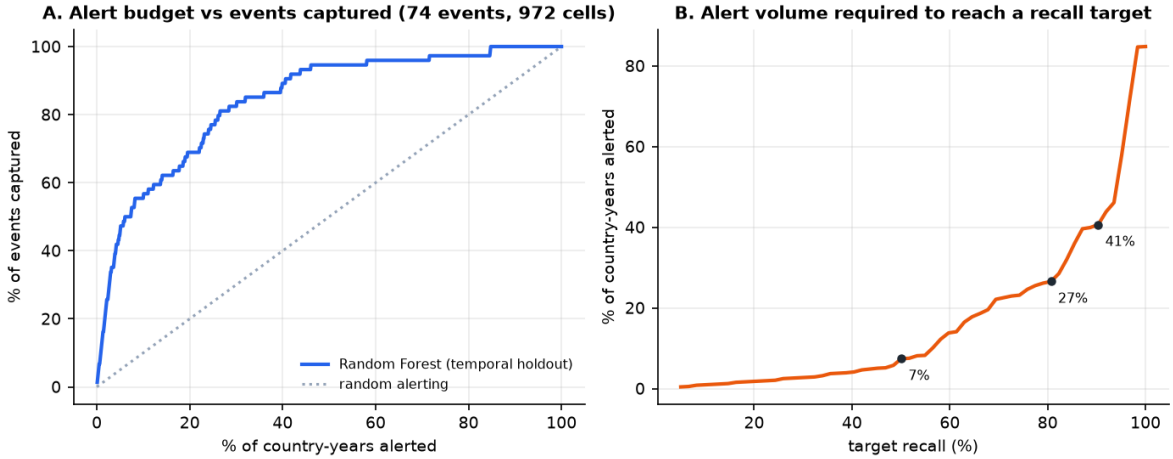


Figure 2: Alert economics. Roughly 6% of country-years must be alerted to reach 50% recall; 28% reaches 80%, against 6% for random alerting.

This is the operationally meaningful result: alerting 6% of country-years captures half of future documented events, roughly eight times the efficiency of arbitrary selection.

Table 7: Leave-country-out performance by region.

Region	n	**events**	**ROC-AUC**	**PR-AUC**
East Asia & Pacific	456	48	0.865	0.396
Europe & Central Asia	912	31	0.826	0.163
Latin America & Caribbean	551	12	0.903	0.381
MENA, Afghanistan \& Pakistan	380	23	0.834	0.220
Sub-Saharan Africa	665	27	0.699	0.073

Where it fails. Skill is lowest in Sub-Saharan Africa and in low-income countries — precisely where surveillance is thinnest and an early-warning system would matter most. The mechanism is label sparsity, not model failure. These strata should not be treated as interchangeable in a deployment decision.

Caveats. The label is documented activity, and a zero conflates absence of disease with absence of publication; all scores describe skill at predicting surveillance-observable events. Text-mined labels inherit extraction error and the extractor has not been validated against a hand-annotated gold set. Country-level aggregation hides the district heterogeneity that matters in India. With 172 positives, intervals are wide ($\approx \pm 0.11$ on the headline PR-AUC) and subgroup estimates are noisy.

Geographic and socio-economic features act largely as reporting-capacity proxies, so no causal or intervention claim follows.

Reproducibility

Scripts `src/00` – `src/08` run in order. Cached payloads in `data/raw`; assembled panel and events in `data/processed`; all reported numbers in `reports/results.json` and `reports/dataset_analysis.json`. Every table and figure here is generated from those files — the report contains no hand-transcribed results.

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